



Society of Petroleum Engineers

SPE-227786-MS

Visual Understanding of Rock Wettability Distribution to Contact Angle Regions and Oil Displacement Patterns in the Silty Sand Reservoir via 2D Pore-Scale Modeling

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227786-MS](https://doi.org/10.2118/227786-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

This study, conducted using a silty sand reservoir rock extracted from a Kuwait-producing oilfield, provides crucial insights into fluid distribution patterns and mineralogy, as well as wettability contact angle preferences. The method used for visual identification not only captures rock physics but also suggests effective oil recovery displacement strategies. Visual identification is presented using 2D image technology and a Scanning Electron Microscope (SEM). Analytical data are presented from electron bombardments and backscattering reflections collected in the BSE detector. These analyses were used to characterize the various surface boundary morphology parameters of the silty sand mineral surfaces, including area, perimeter, mineral surface angles, and pore counts. Further necessary morphological measurements are length, width, and the static wettability contact angle between pore and grain spaces across the sample. The pore-wall wettability contact angle is accurately quantified by acquiring pore-scale images and original fluid distribution and capturing all counted pores. The 2-D approach has identified eighteen regions of fluid distribution patterns, of which about 38% are newly detected wettability regions, considering the geometrical contact angle mirror effect. The morphological statistical pore parameters are counted for 1,877 objects of pores at an X40 magnification level, with pore areas ranging from 7.8 μm^2 to 713,905.3 μm^2 . This magnification level captures the most significant wettability-measured feature from the silty sand rock. The overall pore-grain wall wettability preference is medium water wet (MWW) with an overall averaged wettability contact angle of 35.35°, which makes the silty sand rock a candidate for secondary water displacement treatments. This research has significant implications for the petroleum industry, particularly in understanding the wettability contact angle distribution within reservoir rock, a critical factor in Al-Bazzaz's models. It will inspire and guide any future crude oil development in this field, opening new possibilities for cost-effective oil recovery strategies during the life cycle of this reservoir.

Keywords: Silty Sand, Wettability Preference, 2D-Digital Contact Angle, Water Wet Fluid Distribution, Oil Recovery

Introduction

The study of wettability, a phenomenon that describes the quantitative and qualitative distribution of fluids on the surfaces of rock pores on both micro and macro scales, significantly enhances our understanding of the movement of fluids, displacement patterns, and fluid distribution within these pores (Sharifigaliuk et al., 2021). Several definitions of digital wettability classifications have been proposed in previous studies, including mixed-wet, intermediate-wet, neutral-wet, weakly water-wet, and weakly to strongly water-wet. Most of these definitions rely on either quantitative measurement of contact angles, θ , and interfacial tension, σ /IFT, using the traditional Amott, USBM, and combined Amott-USBM methods or qualitative measurements on core samples through macroscopic experiments (Abdallah et al., 2007). The benefit of the first approach is to define the wetting preference of rocks, considering the geometry and shape of the pore structure in porous media. As for the macroscopic measurements on the fragment and core samples, they are not considered entirely suitable for determining the specific wettability of most reservoir rocks. However, the macroscopic approach provides reliable quality information on wettability preference. Therefore, the surface morphology of the pore spaces, the spatial distribution of minerals, and the deposition of organic materials on the surface of minerals prompted us to study the various types of reservoir rocks, including the silty sand rock, the focus of our study, because of its importance in determining and understanding the characterization of the rock wettability (Almudhhi et al., 2021).

Silty sand is a clastic sedimentary rock consisting primarily of silt and a low clay mineral content (Fig. 1). It is distinguished by its fine and coarse sand grains, with wide particle size variations of silts and clays, and a lack of fission. Silty sand usually contains sand particles (0.05 mm - 0.2 mm), fine silt particles (0.002 mm – 0.05 mm), and clay particles (<0.002 mm). Reservoirs containing silt in their rocks can complicate their characterization and lead to misinterpretations in identifying reservoirs or hydrocarbon-producing zones, as well as in determining clay volume, porosity, and fluid saturation (Al-Hitti et al., 2005; Gaafar et al., 2016). The Silty sand reservoir is a fresh rock fragment selected from a Kuwaiti reservoir core in its native state, containing hydrocarbons. This sample is selected due to the complexity of oil recovery using conventional methods.

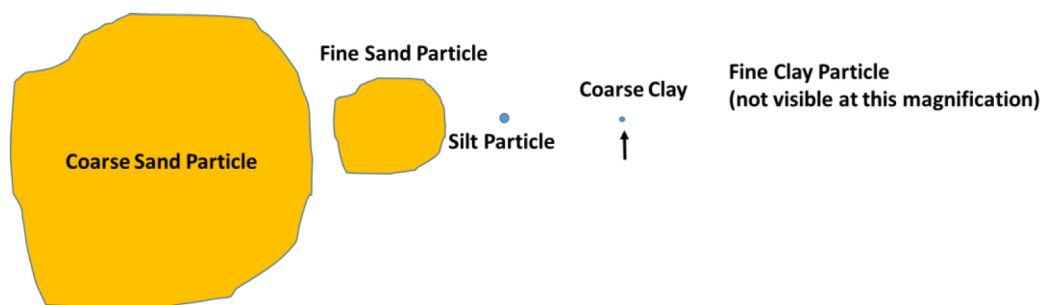


Figure 1—The Silty Sand Reservoir Complexity of Different Particle Sizes is Considered a Significant Source of Wettability Heterogeneity between Coarser Sand Particles (0.2 mm), Silt Particles (0.002 mm), and Fine Clay Particles (<0.002 mm).

Wettability, the preference of a solid in contact with one fluid over another, is a crucial concept in the hydrocarbon upstream regions. Reservoir rocks, representing the solid surface, and crude oil and brine, the fluid phase, are at the heart of this research. The heterogeneous nature of wettability in porous media and oil reservoirs, where the rock's surface exhibits a non-uniform molecular affinity for the contact fluids in different regions, underscores the significance of this study (Paterson et al., 1998).

Several studies have shown the effect of various parameters on the distribution of fluids within porous media and the wettability of rocks. Taha et al. (2001) described the variation in wettability as being accompanied by changes in waterflood efficiency, and that the wettability of the reservoir changes from water-wet, low on structure near the oil/water contact, to mixed or neutral wet behavior higher on structure. Aspenes et al. (2003) investigated in situ wettability distribution and wetting stability of moderately water-wet and nearly neutral-wet outcrop chalk cores at elevated temperatures. The study demonstrated that multidirectional crude oil flooding resulted in uniform wettability distributions throughout the reservoir. A dome-shaped axial wettability distribution exhibited more water-wet conditions at the center of the cores, and the variation was only significant near the inlet end. Arsalan and Nguyen (2016) introduced an approach to characterize the mixed-wet reservoir using Inverse Gas Chromatography (IGC). The study reveals that the Lifshitz-van der Waals component remains unchanged with variations in the wetting state of the system under study, and the deposition of organic deposits destroys the strongly polar sites initially present on the reservoir rock. Cobos et al. (2021) evaluated wettability alteration, and the results confirmed that supplying fresh crude oil during the aging procedure to a heterogeneous limestone plug causes the most significant wettability change. Yongfei et al. (2021) studied the distribution and influencing factors of the remaining oil in water flooding at the pore scale. The results showed that the complex pore structure would lead to the trapping of the remaining oil, and the phase circulation phenomenon can be observed in the remaining oil.

Additionally, the results reveal distinct wettability distributions in trapping patterns and front instability. Jonas et al. (2022) investigated the influence of spatially heterogeneous wettability distributions on infiltration into porous media. The results demonstrate that water repellency in porous media can occur for non-hydrophobic contact angles, and that the spatial distribution of wettability has a significant influence on water repellency and water infiltration dynamics. Additionally, the study reveals that even small amounts of these layers in the correct location can induce water repellency, as the pore throats are among the last parts of a rock to dry, and the dependence of the contact angle on water content, particle geometry, and surface roughness. Himanshu and Kishore (2022) developed a mechanistic model to simulate low salinity waterflood in a carbonate reservoir. The study reveals that waterfloods for oil recovery from carbonate reservoirs are often ineffective due to their oil-wet nature, despite holding substantial oil reserves. However, oil recovery from oil-wet carbonate reservoirs can be increased by wettability alteration using low-salinity brines having carefully modified ionic compositions. Most open-source studies emphasized the importance of fluid distribution in porous media and its effect on initial wettability and the extent of wettability alteration in enhanced oil recovery.

According to Al-Sayegh et al. (2023), this study employs a unique approach, utilizing a silty sand reservoir extracted from a Kuwaiti oilfield that produces oil and features a heterogeneous pore structure. Fragments from the core saturated with crude oil are used in the Scanning Electron Microscope (SEM) chamber and backscattered electron detector (BSE), as described by Al-Bazzaz et al. (2007). The 2D image technology is used to acquire pore-scale images, characterize the morphology parameters of pore and grain spaces across the sample, and measure fluid distribution. Additionally, it allows for counting all pores and determining their pore-wall wettability contact angle. This innovative method aims to understand fluid distribution patterns and wettability preferences by visual identification, capturing rock physics, and suggesting the effective displacement of oil recovery.

Fluid Distributions Based on Wettability Preference

Immiscible fluid displacement patterns in porous media have great interest during oil production and recovery processes, where different parameters directly govern the displacement mechanisms and in-situ fluid distributions within the pores of rocks, including contact angle, wettability, viscosity of fluids, capillary pressure, and gravity forces (Albazzaz et al., 2019; Al-Haleem et al., 2019). In previous work, eighteen patterns for fluid distributions were identified based on the strength of wettability preference and contact angle (Al-Ajaj et al., 2022). These patterns are shown in Table 1. These patterns begin with the water

region being 100% free of oil, then transition to a strongly-water-wet (SWW) state, where pores prefer the spreading of water over oil. This region does not need enhancement because the oil can flow naturally through the primary production. In medium water wet (MWW), the water is dominant in this region; however, a small amount of oil can contact the pore walls. The best displacement mechanism in this region is secondary Water flooding because the pore walls prefer water wetness, which allows the existing amounts of oil to leave their positions and be replaced by water droplets due to attraction forces. The weak-water-wet (WWW) region is the fourth fluid distribution pattern of the reservoir. It can be described as a mixed wettability region with a slight preference for water. Low-salinity water flooding is an effective option for oil displacement in this region, as fines migration during pore sweep can partially block pore space at the walls, diverting the injected water into unswept pores and thereby improving oil recovery. Another region is a mixed wettability zone between the water-wet and oil-wet regions of the rocks, with no preferential wettability for either fluid. The fluid distribution in this region exhibits equal contact intensities between the oil and water on the one hand and reservoir fluids and rocks on the other. Low-salinity water floods, such as those with low heat or smart water, are suggested treatments for enhancing rock wettability in the water and improving oil recovery in this region (Ahmadi et al., 2020). The sixth region is a weak-oil-wet (WOW) zone, where the reservoir rocks slightly prefer an oil-wet condition. Chemical-enhanced oil recovery flooding, or surfactant water floods with low heat, is the recommended process for oil displacement in this region due to the complex phase behavior in the pore space, which can be attributed to the intensity of the contact between the oil and water phases. The seventh fluid distribution pattern in reservoir rocks is described as medium-oil-wet (MOW). In this region, the pore walls are more preferential towards oil wetness and less preferential towards water wetness. A polymer displacement water flood augmented with temperatures between 100 °C and 300 °C is a practical suggestion for oil recovery, as it increases surface activity and water viscosity at elevated temperatures, thereby reducing early fluid breakthrough and improving sweep efficiency. The fluid distribution in the penultimate region exhibits a strong oil-wet (SOW) characteristic, where the pore walls tend to be solidly preferential towards oil wetness and non-preferential towards water wetness. Oil recovery in this region is challenging because the most effective recovery mechanism involves aggressive, unconventional, and costly techniques, such as water displacement flooding with temperatures exceeding 300°C. Finally, the oil-wet region (AOW) can be described as the residual oil saturation region at the end of the water, characterized by immobile oil attached to the pore walls (Al-Ajaj et al., 2024).

Table 1—presents the fluid distribution patterns based on the SEM-BSE measured contact angle wettability preference, suggesting an oil displacement mechanism for each pattern in the 2D approach according to the Al-Bazzaz classification (Al-Ajaj, 2022).

Class #	Angle (θ°)	Wettability Preference	Wettability Preference Abbreviation
1	$\theta^\circ = 0^\circ$	Absolute Horizontal Water Wet	AHWW
2	$0^\circ < \theta^\circ < 30^\circ$	Strongly Water Wet	SWW
3	$30^\circ < \theta^\circ < 60^\circ$	Medium Water Wet	MWW
4	$60^\circ < \theta^\circ < 90^\circ$	Weak Water Wet	WWW
5	$\theta^\circ = 90^\circ$	True Vertical Mixed Wet	TVMW
6	$90^\circ < \theta^\circ < 120^\circ$	Weak Oil Wet	WOW
7	$120^\circ < \theta^\circ < 150^\circ$	Medium Oil Wet	MOW
8	$150^\circ < \theta^\circ < 180^\circ$	Strong Oil Wet	SOW
9	$\theta^\circ = 180^\circ$	Absolute Horizontal Oil Wet	AHOW
10	$\theta^\circ = 180^\circ$	Absolute Horizontal Oil Wet	AHOW
11	$180^\circ < \theta^\circ < 210^\circ$	Strong Oil Wet	SOW
12	$210^\circ < \theta^\circ < 240^\circ$	Medium Oil Wet	MOW
13	$240^\circ < \theta^\circ < 270^\circ$	Weak Oil Wet	WOW
14	$\theta^\circ = 270^\circ$	True Vertical Mixed Wet	TVMW
15	$270^\circ < \theta^\circ < 300^\circ$	Weak Water Wet	WWW
16	$300^\circ < \theta^\circ < 330^\circ$	Medium Water Wet	MWW
17	$330^\circ < \theta^\circ < 360^\circ$	Strong Water Wet	SWW
18	$\theta^\circ = 360^\circ$	Absolute Horizontal Water Wet	AHWW

The two-dimensional technology has enabled the capture of the mirror effect and included nine other regions reflecting complementary wettability contact angles that were absent in conventional trending models, as shown in Figure 1 (Al-Bazzaz et al., 2018).

The 2-D approach doubles the fluid distribution pattern to eighteen regions, as summarized in Fig. 2 and Table 1. It proposed dividing the wetted contact angle of pore geometry into trapezoidal-like angles where the initial and terminal sides of the angle are measured. There are two proposals to explain the congruence of the angle with its counterpart in the mirror effect; the first proposal is that the angle coincides with its initial side, while the second proposal interprets the congruence of the angle with its mirror image in the terminal side. For example, if the initial side of pattern WWW is $\Theta = 60^\circ$, the terminal side of the following mirror is $\Theta = 300^\circ$, while in the second proposal, the terminal side of pattern WWW will be $\Theta = 240^\circ$, as shown in Fig. 2.

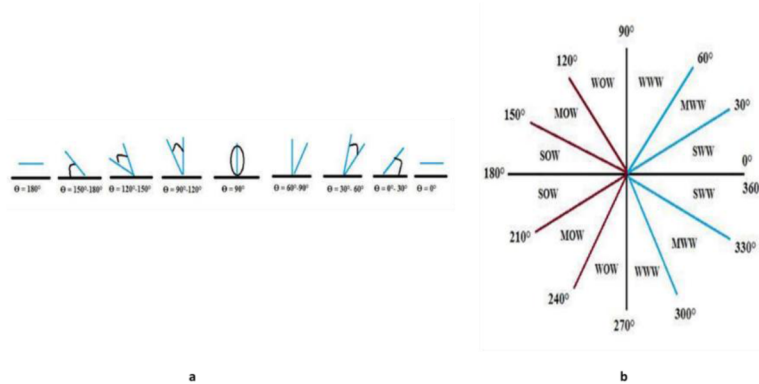


Figure 2—(a) Conventional 1D Conceptual Fluid Distributions on Silty Sand Minerals are Based on Average Measured Digital Contact Angles (Θ°) that will Dictate the Overall Reservoir Rock Sample Wettability Preference, showing 9 wettability Preference Classes between 0° - 180° . (b) Unconventional 2D Conceptual Fluid Distribution on Silty Sand Minerals, showing the Mirror Effect and Doubling Wettability regions to 18 Classes between 0° - 360° , including the Quadrantal Angles (0° , 90° , 180° , 270° , and 360°).

The wettability contact angles (θ°) are all accurately and successfully captured in this study. Complementary and supplementary angles, including adjacent angles of 0° , 90° , 180° , 270° , and 360° (Fig. 2), will be referred to as the missing angles of the wettability contact angles, which were previously absent from conventional knowledge due to their past difficulty in assessment. The straight angle (θ°) 0° , Absolute Horizontal Water Wet angle (AHWW), the right angle (θ°) 90° , True Vertical Mixed Wet (TVMW), the straight angle (θ°) 180° , Absolute Horizontal Oil Wet (AHOW), the reflex angle (θ°) 270° , True Vertical Mixed Wet (TVMW), and the complete angle (θ°) 360° , Absolute Horizontal Water Wet (AHWW) are all known as quadrantal angles. Quadrantal angles are the terminal side of the angle that lies exactly on either the x-axis or the y-axis. Measuring and understanding these quadrantal angle values is crucial for forecasting the oil recovery potentials and treatment programs. Measurements of these angles are now possible with 2D digital image capture using powerful microscopes and advancements in Artificial Intelligence models. 2D digital technology utilizes trigonometric functions to solve wettability contact angle equations.

Methodology

Nanometer, micrometer, and millimeter-sized pores coexist within a single piece of rock. In this case study, an Artificial Intelligence (AI) model is developed that encompasses all pore and grain feature morphological dimensions of the rock microstructure, representing the geometrical properties of all grain minerals and pores, as well as their associated wettability contact angle properties (Al-Sayegh et al., 2023). A two-dimensional (2D) approach is proposed that integrates several digital rock images of different resolutions

within the framework of pore network modeling, as proposed by [Al-Bazzaz et al. \(2007\)](#). In this study, we have used two dimension scattered electron microscopy with back scattered emission (2D-SEM-BSE) technology to understand the complex effect of wettability heterogeneity represented by contact angle classification regions at the pore scale by capturing an image from two scales at X40 (mm scale) ([Fig. 3](#)) and X101 (μm scale) ([Fig. 5](#)) of the rock sample. The purpose of the two suggested magnification scales is to validate the AI model.

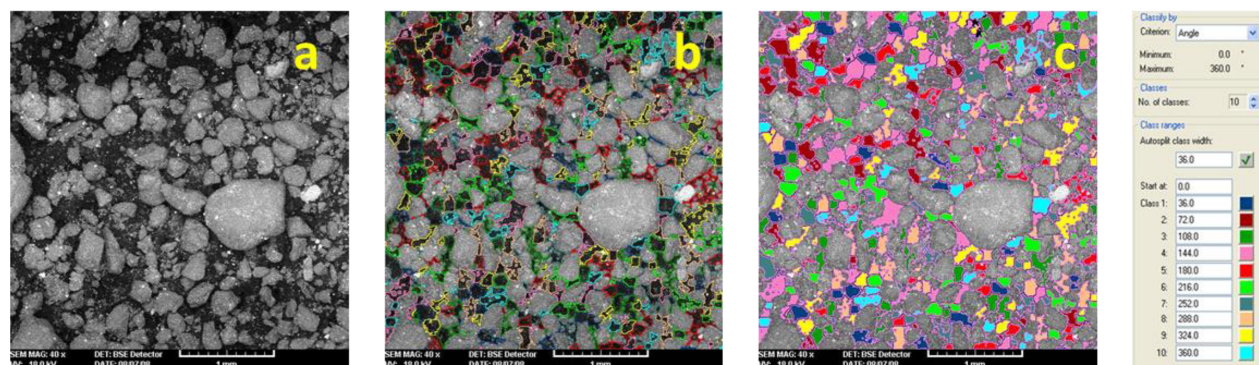


Figure 3—(a) The Captured 40X Magnification SEM Image of the Kuwaiti Silty Sand Core Sample Showing Mineral Different Particle Sizes. (b) Shows the boundary between Different Grain Particle sizes and Pore Spaces where the Contact Angles are Measured. (c) Pore Filled with Oil Constructed by 3 Grains Particles Architecture (Coarse Sand, Silt, and Fine Clay).

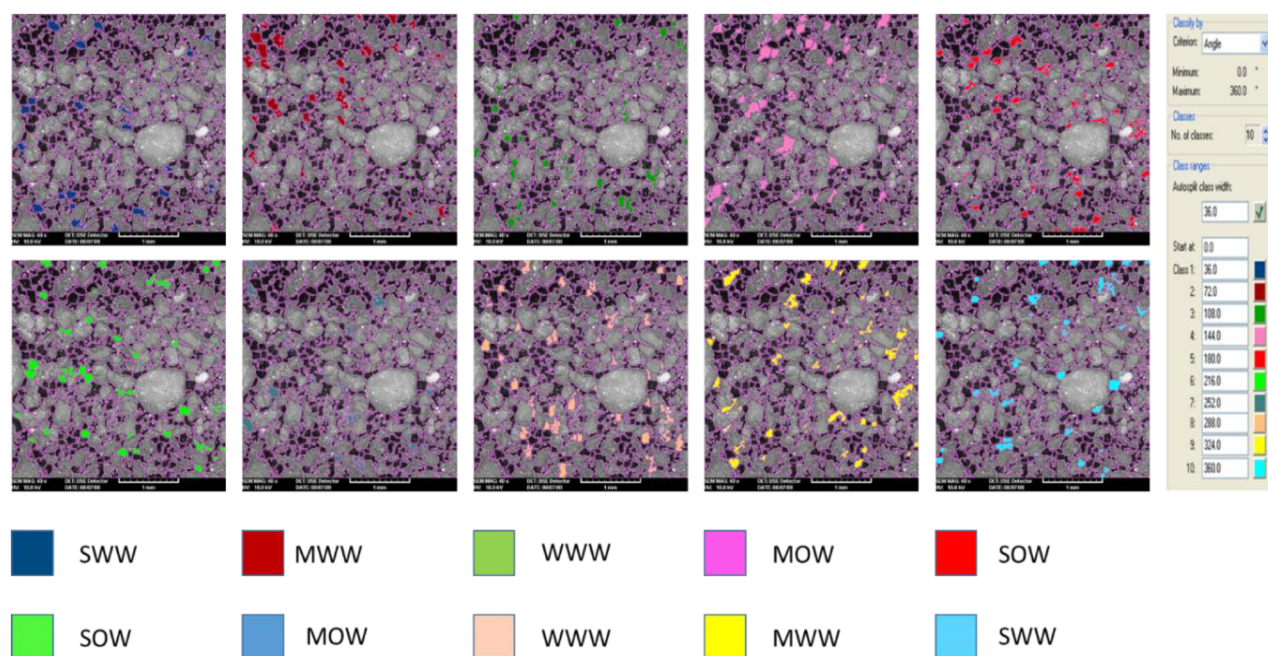


Figure 4—Mapping Wettability Preferences inside the Silty Sand 40X SEM image, Displaying the 10 Color Distribution of Pore Classifications.

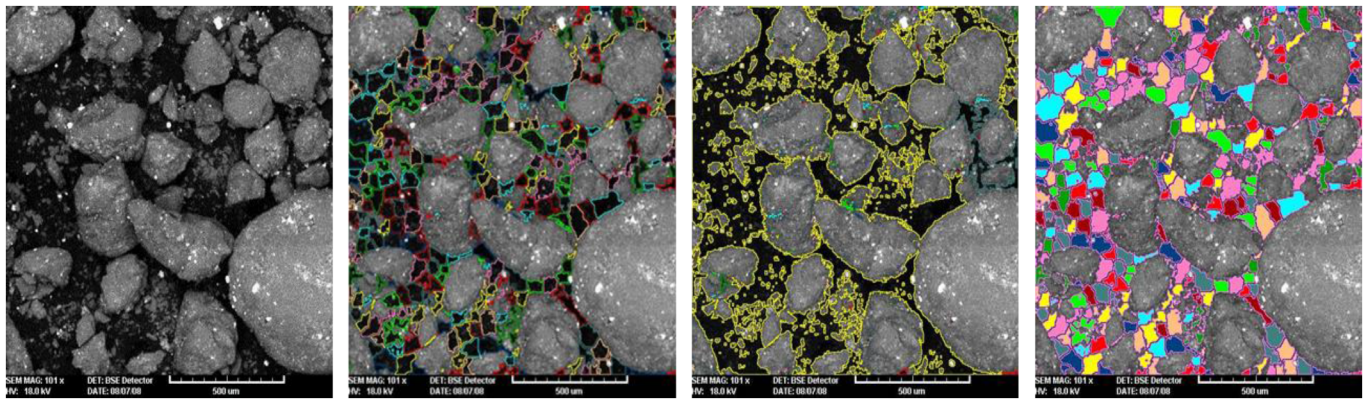


Figure 5—The SEM image of the core sample shows the pore boundary and grain boundary at a magnification of 101X. These images and Data Are for validating the k-means Algorithm Model.

The captured images were scanned using a backscattered electron detector (BSE) available in the digital electron microscope, which can accurately measure several morphological parameters of pore and grain spaces, including pore area, width, and length, as well as pore roundness. Then, wettability is classified by using the concept of 2D geometrical contact angle to describe wettability attributes in the oil recovery models (Fig. 4). The study utilized the measured static contact angle θ° to determine the wettability distribution in the k-means clustering algorithm, which is an unsupervised machine learning algorithm for data grouping that does not require labeled data for training (Table 3). The static contact angles ranged from 0° to 360° . The 2D-SEM-BSE technique is designed for non-destructive core pore imaging, a practice considered a fast, iterative, and robust technique in digital wettability measurement technology (Al-Sayegh et. Al., 2023).

Silty sand, rock fragment samples are selected for image capture (Figures 3, 4, and 5). 2D pores are captured using a big data quantification framework and then morphologically analyzed and modeled using unconventional pore-grain big data statistical artificial intelligence techniques, such as the k-means clustering algorithm (Tables 2 and 3). Determining the wettability of hydrocarbon silty sand (silt, sand, and clay particles) reservoirs is complex due to grain size discrepancy (Fig. 3). The different particle sizes are considered a significant source of wettability complexity between the sand particles (0.2 mm) and the silt particles (0.002 mm). Additionally, there is heterogeneity in mineral content at different scales (Fig. 5). The complexity of morphology features and pore geometry complicates the measurement of contact angles. Hence, emerging technologies and unconventional approaches are necessary to better understand and characterize the distribution of silty sand rock wettability. Two techniques emerged for obtaining the basic morphology features of pores: taking images using scanning electron microscopy (SEM) and a backscattered electron scattering imaging (BSE) system. Eight parameters were detected associated directly with the actual captured pores from SEM-BSE Images, which are Pore Counts, Pore Area (A) in μm^2 , Pore Perimeter (P) in μm , Pore Length (L) in μm , Pore Width (W) in μm , Pore Roundness unit less, W/L aspect ratio unit less, and average Pore Contact Angle (θ) degrees.

Table 2—This Dataset is Eliminated from the Quadrantal Angles, 0°, 90°, 180°, 270°, and 360° for this Model. Resulting in an Overall Weak-Oil-Wet WOW = 108.7°, so the Recommended Oil Recovery Treatment is ASP Surfactant Flooding.

Cluster Class	# of Objects	Area (μm) ²	Perimeter (μm)	Length (μm)	Width (μm)	Roundness	Aspect Ratio	Angle (θ°)
1	2,526	28.95	1.40	3.75	3.26	0.44	0.49	0.06
2	255	12.38	1.93	1.10	0.71	0.05	0.07	2.08
3	525	13.82	2.15	1.57	0.81	0.09	0.18	7.64
4	261	21.45	3.32	1.49	0.88	0.05	0.07	4.36
5	312	13.64	2.23	1.25	0.81	0.06	0.08	7.09
6	210	15.09	2.51	1.15	0.70	0.04	0.06	5.53
7	210	21.00	3.22	1.37	0.87	0.05	0.06	7.11
8	322	19.41	2.72	1.39	0.92	0.06	0.08	12.29
9	701	38.96	3.95	2.32	1.24	0.12	0.22	29.53
10	202	12.51	2.09	0.95	0.58	0.03	0.04	7.65
11	271	11.29	1.81	0.99	0.64	0.04	0.06	12.64
12	240	18.09	3.01	1.38	0.77	0.05	0.07	12.71
Σ	6,033	226.59						108.68

Digital styles were used in this study to group the measured contact angle data (Table 3). This AI unsupervised machine learning (k-means) technology is limited to a maximum of ten different colored classes or ten clusters. Each cluster has a fixed color (Figures 3-5). Therefore, the measured contact angle data can be equally spanned between 0° and 360° degrees over 10 clusters, with each cluster spanning 36° degrees, as shown in Fig. 3. These colors will clearly distinguish the images captured using the two-dimensional technology and assign a specific color to each cluster of measured contact angles, making it easier to assess the intensity of each cluster visually.

Table 3—Wettability Preference is Based on the Number of Static Contact Angle Distributions Measured in the Silty-Sand Rock Sample using the k-means clustering Algorithm 40X (μm).

Degrees	Cluster_1	Cluster_2	Cluster_3	Cluster_4	Cluster_5	Cluster_6	Cluster_7	Cluster_8	Cluster_9	Cluster_10
0	0	1138	0	0	0	7910	10	0	0	0
0-30	0	34	7	0	0	0	33	0	1	7
30-60	0	173	8	16	0	0	46	0	0	2
60-90	0	0	7	32	0	0	45	0	1	2
90	0	0	0	664	0	0	9	0	0	0
90-120	0	0	11	47	0	0	36	10	1	5
120-150	0	0	14	175	0	0	27	63	0	3
150-180	0	0	13	0	0	0	10	64	1	6
180	0	0	0	0	0	0	0	10	0	0
180-210	0	0	12	0	12	0	0	53	4	5
210-240	136	0	9	0	17	0	0	44	1	3
240-270	33	0	8	0	37	0	0	6	2	1
270	350	0	0	0	2	0	0	0	0	0
270-300	34	0	4	0	37	0	0	0	2	3
300-330	154	0	5	0	39	0	0	0	0	1
330-359	48	0	11	0	41	0	0	0	2	1
360	1	0	0	0	0	0	0	0	0	0

Results and Discussions

In this study, 2-dimensional technology is used to characterize the morphology of the Silty Sand rock from the Kuwait reservoir. Two steps were used for the analysis: firstly, the image was captured at two magnifications: X40 and X101. Different clusters have been identified based on wettability contact angles and wettability preferences. Second, the imaging process includes measuring area, pore counts, frequencies

at each contact angle, and several other morphological parameters of pore and grain spaces. In 2D SEM-BSE imaging techniques, the morphological features of each pore are counted and characterized separately, and then modeled by combining their overall appearances to quantify the contact angle (Al-Bazzaz et al., 2007).

2D Image Capturing

Two X40 (mm) and X101 (μm) SEM magnifications are studied. The magnification of 40X is sufficient for capturing the rock sample, representing the millimeter scale. A set of images has been investigated for the matrix pore-body, pore-grain boundary, and pore classification, as shown in Fig. 3. The captured images accurately depict the heterogeneity of grains in this type of rock, as well as the large number of pores that appear at this magnification. Also, the captured SEM images showed colored contact regions between the grain wall and the pores in the millimeter pore system. As for the image in which ten different colors appear, it reflects all the possible classes of fluid distribution around the matrix according to the contact angles and the wettability preference.

Figure 4 shows ten SEM images of fluids distributed according to their contact angle and wettability preferences. All images represent a specific class map defined by a previously specified contact angle in the digital software, illustrating the intensity of fluid distribution within this class. Each image covers the pores that lie within 36° of the contact angle, and with these ten images, we cover all 360° of the contact angles. Several classes emerged based on the counting and characterization of each pore morphological feature. The data generated will be used to estimate the averaged pore wettability contact angles per class, as well as the overall wettability contact angle of the studied silty sand rock.

The second magnification of 101X is the micrometer scale of pore spaces. A set of images has been investigated for the matrix pore-body, pore boundary, grain boundary, and pore classification, as shown in Figure 5. With a magnification of about double, we notice that the image appears more distinct regarding fluid distribution and pore classification. Despite this, the number of pores appeared to be less than the previous magnification, due to the deduction of part of the area, and some of these pores were absent. So, the last magnification of 40X will be adopted in the calculations, considering this magnification of 101X for comparison and confirmation.

Pore morphology and Wettability Modelling

Two-dimensional imaging, a novel approach, was employed to investigate the morphology of silty sandstone. More than ten morphological parameters of pore and grain spaces have been measured from BSE-captured images using SEM based on statistical pore number object counts and geometrical parameters. These include pore counts, pore area (μm^2), pore perimeter (μm), pore roundness (unit-less), pore width (μm), pore length (μm), and pore aspect ratio (Length/Width) (unit-less). The data generated will be used to estimate the averaged pore wettability contact angles per cluster and the overall sample wettability contact angle. Two different magnifications, X40 and X101, have been set to accurately evaluate wettability, wetting contact angle, and wettability preference, as shown in Tables 2, 3, and 6. Table 4 presents the statistical pore morphology parameters at 40X magnification, based on the counting of 1,877 pores. The pore area ranges from 7.8 to 71,390.3 μm^2 , indicating the heterogeneity of the wettability contact angle feature of the silty sand rock, which is 35.35° , resembling a strong-water-wet (SWW) condition. A more significant observation has been noticed at a magnification of X101 with a total number of objects of 508, as shown in Table 5. The Pore area ranges from 7.8 - 9,945 μm^2 with a profound wettability contact angle of 147.2° , which resembles Medium-Oil-Wet (MOW). This observation suggests that more contact angle measurements will result in a greater water-wet preference, while fewer contact angle measurements tend to favor a more oil-wet preference.

Table 4—Statistical morphological parameters of pore and grain spaces of silty sand rock at X40 magnification.

Object	Class	Area (μm ²)	Perimeter (μm)	Length (μm)	Width (μm)	Roundness	Aspect Ratio	Angle (θ°)
All objects								
Objects #	1877/1877							
Mean		416.38	20.825	5.915	4.547	1.091	1.268	35.356
Deviation		16489.6	613.271	41.999	39.961	1.887	0.487	79.993
Minimum		7.804	0	2.793	2.793	1	1	0
Maximum		713905.3	26366.54	1751.17	1709.35	77.492	5	357.957

Table 5—Statistical morphological parameters of pore and grain spaces of silty sand rock at X101 magnification.

Object	Class	Area (μm ²)	Perimeter (μm)	Length (μm)	Width (μm)	Roundness	Aspect Ratio	Angle (θ°)
All objects								
Objects #	508/508							
Mean		1415.18	145.51	48.876	32.917	1.736	1.554	147.166
Deviation		1706.63	129.649	39.592	27.53	0.852	0.546	118.95
Minimum		7.804	0	2.793	2.793	1	1	0
Maximum		9944.99	576.88	165.79	106.491	5.801	5.597	358.389

Table 6—The X40 magnification of Class Wettability Contact Angle Distribution Inside the Silty Sand Reservoir Rock Sample Showing Detailed 18 Classes Wettability Preference, Type of recovery, and Potential % oil Recovery. This Analysis Hides the Effect of Quadrantal Angles, 0°, 90°, 180°, 270°, and 360°. Additionally, the Overall Medium-Water-Wet MWW is 35.35°, indicating that the Recommended Oil Recovery Treatment is Secondary Water Flooding.

Angle Class	Contact Angle Range	Pore Count/Class	% of Pores	Class Averaged Wettability Contact Angle	Wettability Preference	% Type Oil Recovery	% Oil Recovery
1	0°	1461	77.84	0°	AHWW	Inconclusive	0%
2	0° - 30°	9	0.48	18.24°	SWW	Primary	12.3%
3	30° - 60°	40	2.13	45.03°	MWW	Secondary	5.6%
4	60° - 90°	13	0.69	72.02°	WWW	Low Salinity Water Flooding	3.3%
5	90°	126	6.71	90°	TVMW	Inconclusive	0.0%
6	90° - 120°	16	0.85	105.74°	WOW	Surfactant Flooding	10.9%
7	120° - 150°	43	2.29	133.67°	MOW	Polymer Gel-High Heat	5.5%
8	150° - 180°	10	0.53	164.27°	SOW	Water Flooding at High Temperatures >300°C	9.4%
9	180°	1	0.05	180°	AHOW	Inconclusive	0.0%
10	180°	0	0.00	180°	AHOW	Inconclusive	0.0%
11	180° - 210°	16	0.85	194.59°	SOW	Water Flooding at High Temperatures >300°C	12.2%
12	210° - 240°	24	1.28	224.98°	MOW	Polymer Gel-High Heat	7.5%
13	240° - 270°	13	0.69	254.16°	WOW	Surfactant Flooding	7.4%
14	270°	49	2.61	270°	TVMW	Inconclusive	0.0%
15	270° - 300°	13	0.69	286.45°	WWW	Low Salinity Water Flooding	9.8%
16	300° - 330°	26	1.39	315.57°	MWW	Secondary	5.9%
17	330° - 360°	17	0.91	346.01°	SWW	Primary	10.3%
18	360°	0	0.00	360°	AHWW	Inconclusive	0.0%
Silty Sand	Overall	1877	100	35.35°	MWW	Secondary	100%

The detected objects of pores were distributed randomly across the contact angle, spanning from 0° to 360°. Figure 6 shows the frequency of pore distribution at the wettability contact angle and the classification of the ten digital clusters (Table 3). The results show that pores fill within all the digital clusters and in all our suggested clusters, including the complementary wettability contact angles (mirror effect). The maximum pore count of 78.3% was within the first cluster at a magnification of X40, as shown in Figure 6. A similar observation was noted at a magnification of X101, with a percentage of 26%.

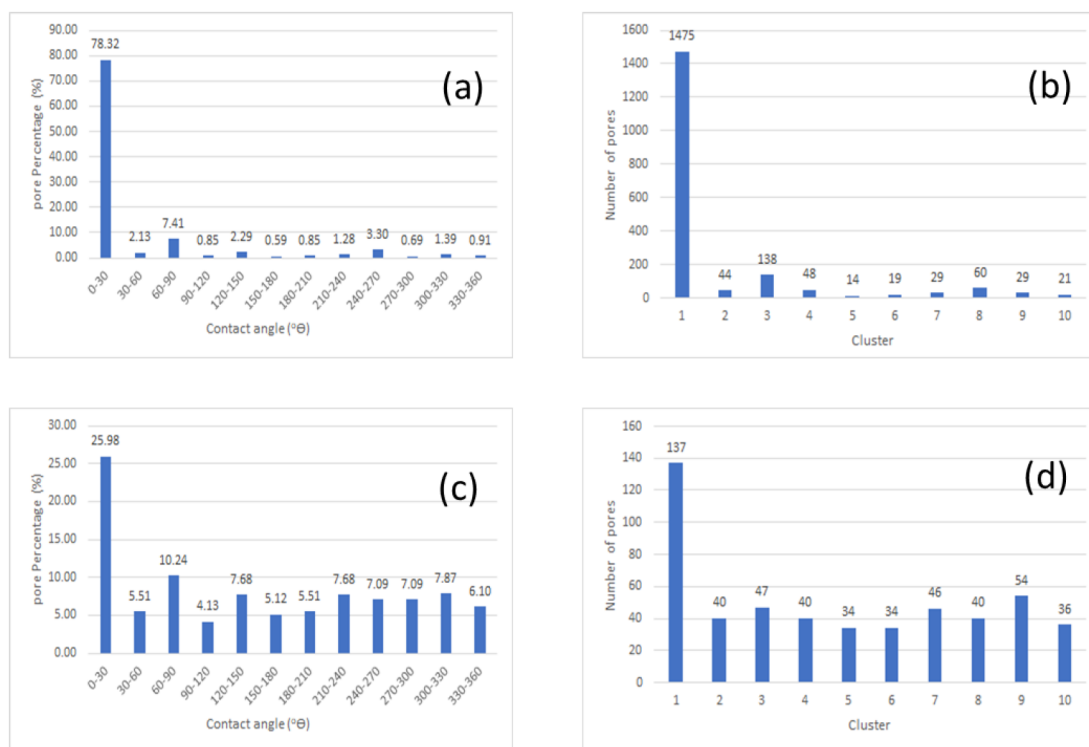


Figure 6—Distributions of Pore Percentage per Degree, Range of 0° - 360°, and Number of Pores Per Cluster in the Silty Sand Rock at (a) and (b) a Magnification of X40 and at (c) and (d) a Magnification of X101.

Based on the measured data using the SEM-BSE 2D technique, the wettability preferences of silty sand reservoir rock have been modeled using a novel k-means clustering algorithm classification, which involves eighteen clusters. Individual wettability preferences have been identified that cover all wetting contact angles and the overall wettability preferences of the reservoir rock sample.

Table 6 and Fig. 7 show the class angle, pore count per class, the average wettability contact angle, and the overall wettability contact angle at a magnification of X40. The model yields the following results: Cluster 1 encompasses 1461 boundary surfaces with 0° contact angles, spanning a 77.84% confidence rate of pores and grain-wall boundaries. The distribution of pores revealed that cluster 1 is a water-rich region and does not contain any oil. The other 22% shows different wettability preferences, including a mixed wettability region between the water-wet and the oil-wet of the rocks, with no preferential wettability for any of the fluids. Cluster five encompasses 126 boundary surfaces at 90° contact angles, spanning a pore percentage of 6.7%. The fluid distribution in this region has an equal wettability preference for oil and water. Low-salinity water floods with low heat, or Smart Water, are suggested treatments for amending rock wettability to water-wet and improving oil recovery in this region. The overall pore-grain wall wettability preference was medium water wet (MWW), with a wettability contact angle of 35.35° at cluster three. The silty sand rock will be an excellent candidate for water displacement developments. The best displacement mechanism in this region is secondary Water flooding because the pore walls prefer water wetness, which allows the existing amounts of oil to leave their positions and be replaced by water droplets due to attraction forces.

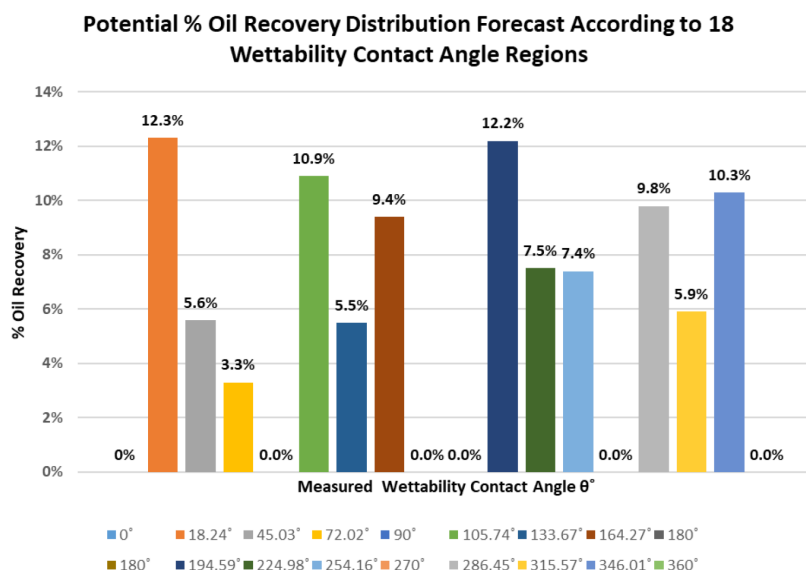


Figure 7—The Oil Potential Recovery Distribution for the entire Silty Sand Reservoir System. According to the Wettability Contact Angle Measurements Using Powerful Visualization, the 2D-SEM-BSE Digital Imaging Technology and k-means Unsupervised Machine Learning Algorithm

In Fig. 7, the Silty sand clusters from 1 to 18 have average values of potential oil recovery, reflecting the measured contact angle. $12.3\% + 10.3\% = 22.6\%$ is the maximum total potential oil recovery for the SWW (Primary Recovery) class-cluster. It is also the highest recovery predicted among the other clusters. The lowest oil production is $5.6\% + 5.9\% = 11.5\%$ of MWW (Secondary Water flooding). In Figure 8, the easy oil classes are 1, 2, 3, 10, 11, and 12, yielding an oil recovery rate of 47.1%, which corresponds to the water-wet region in this silty sand reservoir. Classes 4, 5, 6, 7, 8, and 9 are located in the oil-wet region of the reservoir, collectively accounting for 52.9% of the total. The quadrantal angles of 0° , 90° , 180° , 270° , and 360° are the missing angles in conventional laboratory measuring techniques. The absence of these missing data can affect the assessment of the overall wettability preference and hence the overall oil recovery in this Silty sand reservoir.

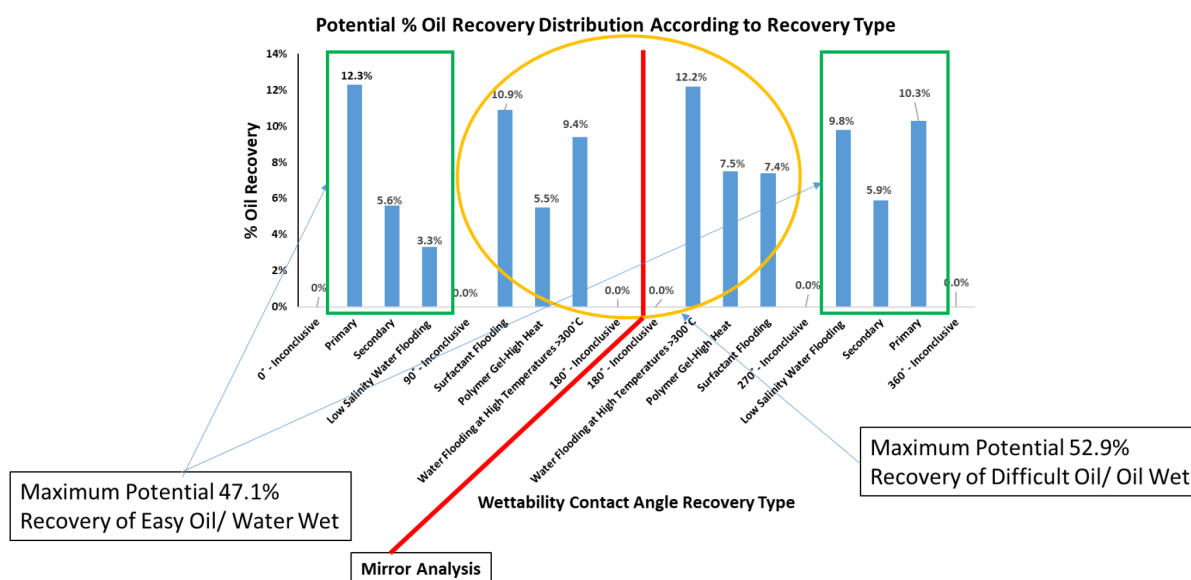


Figure 8—Detailed Potential Oil Recovery Distribution Forecast with 18-Cluster Classes that Suggest the Silty Sand Wettability Contact Angle Preference along the Percentage and the Recovery Type.

The maximum recovery has been modeled using both 18 and 12 contact angle clusters, as well as the reflecting complementary wettability contact angles (mirror effect), as described by Al-Bazzaz's analyses. Each class gathered with its reflecting complementary wettability contact angles. Figure 9 shows the pore count distribution percentage for wettability contact angles. This figure includes the detected complementary wettability contact angles or mirror effect, excluding the pore count at quadrantal-degree contact angles. The maximum peak was at 90-degree contact angles with a total pore percentage of 48.3%. Additionally, the figure indicates that the oil-wet and water-wet regions in the conventional trending model (without the mirror effect) account for approximately 62% of the total studied areas. In comparison, newly detected wettability regions have been added using the SEM-BSE 2D technique, representing about 38% after considering the mirror effect.

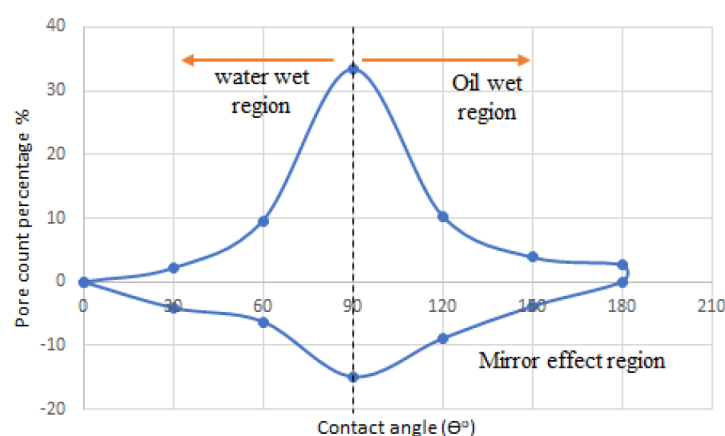


Figure 9—Pore Count Distribution Percentage Concerning Wettability Contact Angles at the Conventional Mirror Model.

Table 7 shows the maximum oil recovery according to Al-Bazzaz's classification for each wetting class pore distribution of the studied rock sample at a Magnification of X40 (Al-Bazzaz et al., 2018). According to Table 7, the total oil recovery in the general water-wet region accounts for approximately 67.2% using the Al-Bazzaz K-means model. The overall pore-grain wall wettability preference is for water-wet conditions, as indicated by the spread between SWW (47.15%), MWW (8.68%), and WWW (11.34%). The remaining overall pore-grain wall boundaries tend towards an oil-wet spread, with mixed wetness of WOW = 15.19%, MOW = 10.69%, and SOW = 60.96%, resulting in percentages of 32.8% according to Al-Bazzaz's model. The silty sand rock will be an excellent candidate for primary recovery and water displacement developments.

Table 7—Total Oil Recovery According to Al-Bazzaz for Each Wetting Class Pore Distribution Available in the Silty Sand Rock Sample at Magnification of X40, Showing the Dominance of Water Wet Region is Strong- Water-Wet (Primary Recovery) at 47.2% out of 67.2%, Yielding 70%. This Analysis Includes All Measured Angles.

Silty Sand Class Wettability Preference	% Oil Recovery	Oil Recovery Dominance	Total % Oil Recovery	Wettability Preference
Class 1 + Class 12 = SWW	47.15%	Primary Recovery	67.2%	Generally Water Wet
Class 2 + Class 11 = MWW	8.68%			
Class 3 + Class 10 = WWW	11.34%			
Class 4 + Class 9 = WOW	15.19%	AFP Surfactant Flooding	32.8%	Generally Oil Wet
Class 5 + Class 8 = MOW	10.69%			
Class 6 + Class 7 = SOW	6.96%			

Table 8 shows the maximum oil recovery according to Al-Bazzaz's analyses. The model predicted the oil recovery for each wetting class pore distribution of the studied rock sample at a Magnification of X40. According to Table 7, the total oil recovery in the general water-wet region accounts for approximately

47.2% using the Al-bazzaz K-means model. The overall pore-grain wall wettability preference is for water-wet conditions, as indicated by the spread between SWW (22.6%), MWW (11.4%), and WWW (13.14%). The remaining overall pore-grain wall boundaries tend towards an oil-wet spread, with mixed wetness of WOW = 18.27%, MOW = 12.92%, and SOW = 21.62%, resulting in a percentage of 52.8% according to Al-Bazzaz's model. The silty sand rock will be an excellent candidate for primary recovery and water displacement at high temperatures ($>300^{\circ}\text{C}$) during development. This analysis excludes the contribution of the quadrantal angles of 0° , 90° , 180° , 270° , and 360° . Hopefully, in the future, more technology will be used to find the missing angles of quadrilaterals. The knowledge of the wettability preference is incomplete without these missing angles and their contribution towards oil recovery.

Table 8—Total Oil Recovery According to Al-Bazzaz for Each Wetting Class Pore Distribution Available in the Silty Sand Rock Sample at Magnification of X40, Showing the Dominance of Water Wet Region is Strong- Water-Wet (Primary Recovery) at 22.6% out of 47.2%, Yielding 48%. This Analysis excludes all Quadrantal Measured Angles, 0° , 90° , 180° , 270° , and 360° .

Silty Sand Class Wettability Preference	% Oil Recovery	Oil Recovery Dominance	Total % Oil Recovery	Wettability Preference
Class 1 + Class 12 = SWW	22.60%	Primary Recovery	47.2%	Generally Water Wet
Class 2 + Class 11 = MWW	11.45%			
Class 3 + Class 10 = WWW	13.14%			
Class 4 + Class 9 = WOW	18.27%	Water Flooding at High Temperatures $>300^{\circ}\text{C}$	52.8%	Generally Oil Wet
Class 5 + Class 8 = MOW	12.92%			
Class 6 + Class 7 = SOW	21.62%			

Conclusions

The SEM-BSE 2D technique shows the unique fluid distribution and the preferred type of silty sand reservoir rocks for wettability. This study concluded that the rocks of this reservoir exhibit various wettability performances, with a dominant preference for mixed water wetting between MWW and WWW, extending to a contact angle of 90° . The latter 90° is usually considered an absent measurement in today's conventional wettability laboratory. Additionally, all difficult quadrantal angles, such as 0° , 90° , 180° , 270° , and 360° that are typically absent in conventional wettability measurement studies, are successfully captured and measured. This quadrantal angles visualization will enhance oil production predictions throughout the reservoir's economic lifespan. The overall pore/grain-wall wettability preference for this silty sand reservoir was medium water wet (MWW), with an average pore-wall wettability contact angle of 35.35° . This assessment is due to the dominant contact angle contribution of the 0° angle, which is available at the Class 1 cluster and is 77.8% abundant in the silty sand reservoir. The significant portions of natural pore voids and grain walls with discrete micro-pores of varying sizes, at a magnification of X40, are fast. Based on k-means modeling of the natural pore-wall boundary system, a water flooding program is proposed to enhance oil potential recovery and improve production, achieving more than 11.45% of the existing oil primary recovery.

Acknowledgments

The authors would like to extend their gratitude to the University of Missouri Science & Technology for their academic support and to the Kuwait Institute for Scientific Research for their imaging technology.

References

1. Sharifigaliuk, H., Mahmood, S.M., Al-Bazzaz, W., and Khosravi, V. Complexities Driving Wettability Evaluation of Shales toward Unconventional Approaches: A Comprehensive Review. *Energy & Fuels* 2021 **35** (2), 1011–1023. DOI: [10.1021/acs.energyfuels.0c03661](https://doi.org/10.1021/acs.energyfuels.0c03661).
2. Wael Abdallah, Jill S. Buckley, Andrew Carnegie, John Edwards, Bernd Herold, Edmund Fordham, Arne Graue, Tarek Habashy, Nikita Seleznev, Claude Signer, Hassan Hussain, Bernard

- Montaron, Murtaza Ziauddin. Fundamentals of Wettability. *Oilfield Review*. June 2007 Oilfield Review **19**(2):44–61.
3. Almudhhi, S., Alostath, M., Al-Bazzaz, W., Sharifigaliuk, H. & Qubian, A. "An unconventional approach in investigating wettability contact angle measurement in shale resources." *Petroleum Science & Technology*, Vol. **40**, Issue 8, Pages 893–936. Published Online: 09 Dec 2021. <https://doi.org/10.1080/10916466.2021.2008969>
 4. Al-Hitti, A.H., Al-Assaf, S.A., Ibrahim, D.S. Reduction of Formation Damage due to Drilling Muds. *Journal of Engineering*, Vol. **11**, Issue 1, Pages 21–32, March 2005, ISSN: 1726-4073 (Print), ISSN: 2520-3339 (Electronic), <https://doi.org/10.31026/j.eng.2005.01.04>
 5. Gamal Ragab Gaafar, Michael Mehmet Eltunbay, Shaharudin B A Aziz, Ehab Najm. Sand-Silt-Clay Evaluation Models: Which One to Use-A Case Study in the Malay Basin. OTC-26771-MS. 2016.
 6. A. Paterson, M. Robin, M. Fermigier, P. Jenffer, J.P. Hulin, Effect of density and spatial distribution of wettability heterogeneities on contact angle, *Journal of Petroleum Science and Engineering*, Volume **20**, Issues 3-4, 1998, Pages 127–132, ISSN 0920-4105, [https://doi.org/10.1016/S0920-4105\(98\)00011-4](https://doi.org/10.1016/S0920-4105(98)00011-4)
 7. Taha M. Okasha, James J. Funk, and Yaslam S. Balobaid. Wettability and relative permeability of lower Cretaceous carbonate rock reservoir, Saudi Arabia. SCA 2001-03. And Okasha, T.M., Funk, J.J., and S.M. Al-Enezi. "Wettability and Relative Permeability of Lower Cretaceous Carbonate Rock Reservoir, Saudi Arabia." Paper presented at the Middle East Oil Show, Bahrain, June 2003. doi: <https://doi.org/10.2118/81484-MS>
 8. E. Aspenes, A. Graue, J. Ramsdal, In situ wettability distribution and wetting stability in outcrop chalk aged in crude oil, *Journal of Petroleum Science and Engineering*, Volume **39**, Issues 3-4, 2003, Pages 337–350, ISSN 0920-4105, [https://doi.org/10.1016/S0920-4105\(03\)00073-1](https://doi.org/10.1016/S0920-4105(03)00073-1)
 9. Arsalan, Naveed, and Quoc P. Nguyen. "Characterization of Mixed Wettability using Surface Energy Distribution." Paper presented at the SPE Improved Oil Recovery Conference, Tulsa, Oklahoma, USA, April 2016. doi: <https://doi.org/10.2118/179706-MS>
 10. Cobos, J.E., Sandnes, M., Steinsbø, M., Brattekkås, B., Søgaaard, E.G., & Graue, A. (2021). Evaluation of wettability alteration in heterogeneous limestone at microscopic and macroscopic levels. *Journal of Petroleum Science and Engineering*, **202**, Article 108534. <https://doi.org/10.1016/j.petrol.2021.108534>
 11. Yongfei Yang, Shaobin Cai, Jun Yao, Junjie Zhong, Kai Zhang, Wenhui Song, Lei Zhang, Hai Sun, Vadim Lisitsa. Pore-scale simulation of remaining oil distribution in 3D porous media affected by wettability and capillarity based on the volume of fluid method. *International Journal of Multiphase Flow* **143** (2021) 103746, ISSN 0301-9322, <https://doi.org/10.1016/j.ijmultiphaseflow.2021.103746>
 12. Jonas Bentz, Ravi A. Patel, Pascal Benard, Alice Lieu, Adrian Hauptenthal, Eva Kroener. How Heterogeneous Pore Scale Distributions of Wettability Affect Infiltration into Porous Media. *Water* 2022, **14**, 1110. <https://doi.org/10.3390/w14071110>.
 13. Himanshu Sharma, Kishore K. Mohanty. Modeling of low salinity waterflooding in carbonates: Effect of rock wettability and organic acid distribution. *Journal of Petroleum Science and Engineering*. **208**, 2022, 109624. ISSN 0920-4105, <https://doi.org/10.1016/j.petrol.2021.109624>
 14. Al-Sayegh, S., Flori, R., Kholosy, S., Al-Bazzaz, W., Al-Saedi, H., Abbas, A., and Qubian, A. Practical imaging applications of wettability contact angles on Kuwaiti tight carbonate reservoir with different rock types. *AIP Conf. Proc.* **2809**, 030010 (2023) <https://doi.org/10.1063/5.0145483>

15. Al-Bazzaz, W.H., and Y.W. Al-Mehanna. "Porosity, Permeability, and MHR Calculations Using SEM and Thin-section Images for Characterizing Complex Maaddud-Burgan Carbonate Reservoir." Paper presented at the Asia Pacific Oil and Gas Conference and Exhibition, Jakarta, Indonesia, October 2007. doi: <https://doi.org/10.2118/110730-MS>
16. Al-Bazzaz, W.H., Al-Mehanna, Y.W., and A. Gupta. "Permeability Modeling Using Neural-Network Approach for Complex Maaddud-Burgan Carbonate Reservoir." Paper presented at the SPE Middle East Oil and Gas Show and Conference, Manama, Bahrain, March 2007. doi: <https://doi.org/10.2118/105337-MS>
17. Albazzaz, W., Almudhhi, S., & Alostath, M. (2019). Investigating wettability contact angle measurement in Kuwaiti heavy oil reservoir and modeling using 2D imaging technologies. *Petroleum Science and Technology*, **37**(6), 694–700. <https://doi.org/10.1080/10916466.2018.1564767>
18. Ayad A. Al-Haleem, Faleh H.M. Almahdawi, and Dhorgham Skban Ibrahim, Suggested Ideas to Manage Produced Water in Southern Iraqi Oil Fields-*Journal of Engineering and Applied Sciences* **14**(6): 1985–1988, 2019.
19. Al-Ajaj, H., Flori, R.; AlSayegh, S.; AlMubarak, H., Al-Bazzaz, W. Investigation of Pore Geometry Wettability Preference in Oolitic Oil Reservoir: Pore Scale Imaging and Modelling Study. 33rd international symposium of the society of core analysts, Austin, Texas, USA, September 2022. DOI: [10.5281/zenodo.7087702](https://doi.org/10.5281/zenodo.7087702).
20. Al-Ajaj, H., Al-Bazzaz, W.H., Flori, R.E., Ibrahim, D.S., Alsayegh, S., and H. AlMubarak. "Identifying Wettability Preference of Dolomite Rock Based on Pore/ Grain Contact Angle Using the SEM-BSE Technique." Paper presented at the Mediterranean Offshore Conference, Alexandria, Egypt, October 2024. doi: <https://doi.org/10.2118/223209-MS>
21. Saeb Ahmadi, Mostafa Hosseini, Ebrahim Tangestani, Seyyed Ebrahim Mousavi, Mohammad Niazi. Wettability alteration and oil recovery by spontaneous imbibition of smart water and surfactants into carbonates. *Petroleum Science* (2020) **17**:712–721.
22. Amir Jahanbakhsh, Omid Shahrokhi, M. Mercedes Maroto-Valer, Understanding the role of wettability distribution on pore-filling and displacement patterns in a homogeneous structure via quasi 3D pore-scale modelling, *Scientific Reports* | (2021) **11**:17847.
23. Al-Bazzaz, Waleed, AlMuddhhi, Salah, and Mohammed AlOstath. "Investigation Wettability Contact Angle Measurement in Kuwaiti Heavy Oil Reservoir and Modeling Using 2D Imaging Technologies." Paper presented at the SPE International Heavy Oil Conference and Exhibition, Kuwait City, Kuwait, December 2018. doi: <https://doi.org/10.2118/193706-MS>
24. Al-Sayegh, Saleh, Flori, Ralph E., Alajaj, Hussain, and Waleed Hussien Al-Bazzaz. "Investigating Pore Body, Pore Throat, Nano-Pore Wettability Preference in Several Unconventional Kuwaiti Carbonate Reservoirs." Paper presented at the SPE/IATMI Asia Pacific Oil & Gas Conference and Exhibition, Jakarta, Indonesia, October 2023. doi: <https://doi.org/10.2118/215371-MS>
25. Sewailan, Mohammad, and Waleed Al-Bazzaz. "Understanding O/G Shale Reservoir Tortuosity by Imaging Characterizations." Paper presented at the SPE Gas & Oil Technology Showcase and Conference, Dubai, UAE, October 2019. doi: <https://doi.org/10.2118/198659-MS>
26. Al-Saedi, Hasan N., Qubian, Ali, Al-Bazzaz, Waleed, and Ralph Flori. "Experimental Study of Low Salinity Water Flooding: The Effect of Polar Organic Components in Low-Permeable Sandstone Reservoir." Paper presented at the International Petroleum Technology Conference, Dhahran, Kingdom of Saudi Arabia, January 2020. doi: <https://doi.org/10.2523/IPTC-19740-MS>
27. Alhuraishawy, Ali K., Bai, Baojun, Wei, Mingzhen, and Waleed H. Al-Bazzaz. "New Insights of Low Salinity Water Flooding in Low Permeability-Low Porosity Clay Rich Sandstone

- Reservoir." Paper presented at the SPE International Heavy Oil Conference and Exhibition, Kuwait City, Kuwait, December 2018. doi: <https://doi.org/10.2118/193744-MS>
28. Alhuraishawy, Ali K., Hamied, Ramzy S., Hammood, Hajir A., and Waleed Hussien AL-Bazzaz. "Enhanced Oil Recovery for Carbonate Oil Reservoir by Using Nano-Surfactant: Part II." Paper presented at the SPE Gas & Oil Technology Showcase and Conference, Dubai, UAE, October 2019. doi: <https://doi.org/10.2118/198666-MS>
 29. Hammood, Hajir A., Alhuraishawy, Ali K., Hamied, Ramzy S., and Waleed Hussien AL-Bazzaz. "Enhanced Oil Recovery for Carbonate Oil Reservoir by using Nano-Surfactant: Part I." Paper presented at the SPE Gas & Oil Technology Showcase and Conference, Dubai, UAE, October 2019. doi: <https://doi.org/10.2118/198662-MS>
 30. W. Al-Bazzaz, A. Al-Mansour, and A. Qubian. "Characterization of Tayyarat Tight Carbonate Extreme Heavy Oil: Comprehensive Recovery Overview." European Association of Geoscientists & Engineers. IOR+ 2025-23rd European Symposium on IOR, Apr 2025, Volume **2025**, p.1–24. DOI: <https://doi.org/10.3997/2214-4609.202531049>
 31. Almudhhi, S., Abdullah, L., Al-Bazzaz, W.H., Alsayegh, S., Alajaj, H., Flori, A. (2023). "Are Natural Fractures in Sandstone Reservoir: Water Wet-Mixed Wet-Or Oil Wet?" *Journal of Engineering Research*, Volume **11** (1B). DOI: <https://doi.org/10.36909/jer.17719>